



MIND

What Is Consciousness, Really?

Maybe our understanding of the concept
itself is fundamentally wrong

BY TIM BAYNE

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WRITE THIS FROM SANTIAGO, nestled between the snow-capped Andes to the east and the Chilean Coastal Range to the west. I am here with about 500 other consciousness researchers for our most important annual meeting. A key theme of this year's gathering is the distribution of consciousness: When does consciousness first emerge in human development? Which animals are conscious, and might it occur in artificial-intelligence systems? And can we construct a consciousness detector, something that might tell us whether, say, babies, bees or bots are members of the "consciousness club"?

These questions are among the most intractable of scientific problems. Consciousness researchers like me have been tackling them for decades. We've made some progress, yet agreement about how to tell whether an organism or machine is conscious remains elusive. Perhaps part of the problem lies with the question itself. Perhaps "consciousness" as a term doesn't really belong to the language of science.

Consciousness researchers have never agreed on what we mean when we use that word. In some disciplines, "consciousness" means "sentience"; in others, "awareness." Still others use it interchangeably with "experience" or "qualia." The muddled terminology makes it hard to tell whether the different areas of consciousness even have a common subject matter.

If appealing to synonyms doesn't work, maybe a better strategy for saying what we mean by "consciousness" is to point to clear, uncontroversial examples of it. In the same way that you might explain what "jazz" means by pointing to the music of Miles Davis or Ella Fitzgerald, so, too, might you explain what "consciousness" means by pointing to familiar experiences—feeling sand between your toes, tasting a ripe strawberry, seeing the sun reflect off a snow-capped mountain peak.

These are all states that it "feels like something" to be in. In contrast, there is nothing that it feels like to be dreamlessly asleep or deeply sedated. In the 1990s philosopher Ned Block, then at the Massachusetts Institute of Technology, introduced the term "phenomenal consciousness" to refer to this state. It's a simple concept that has played a central role in the past three decades of consciousness sci-

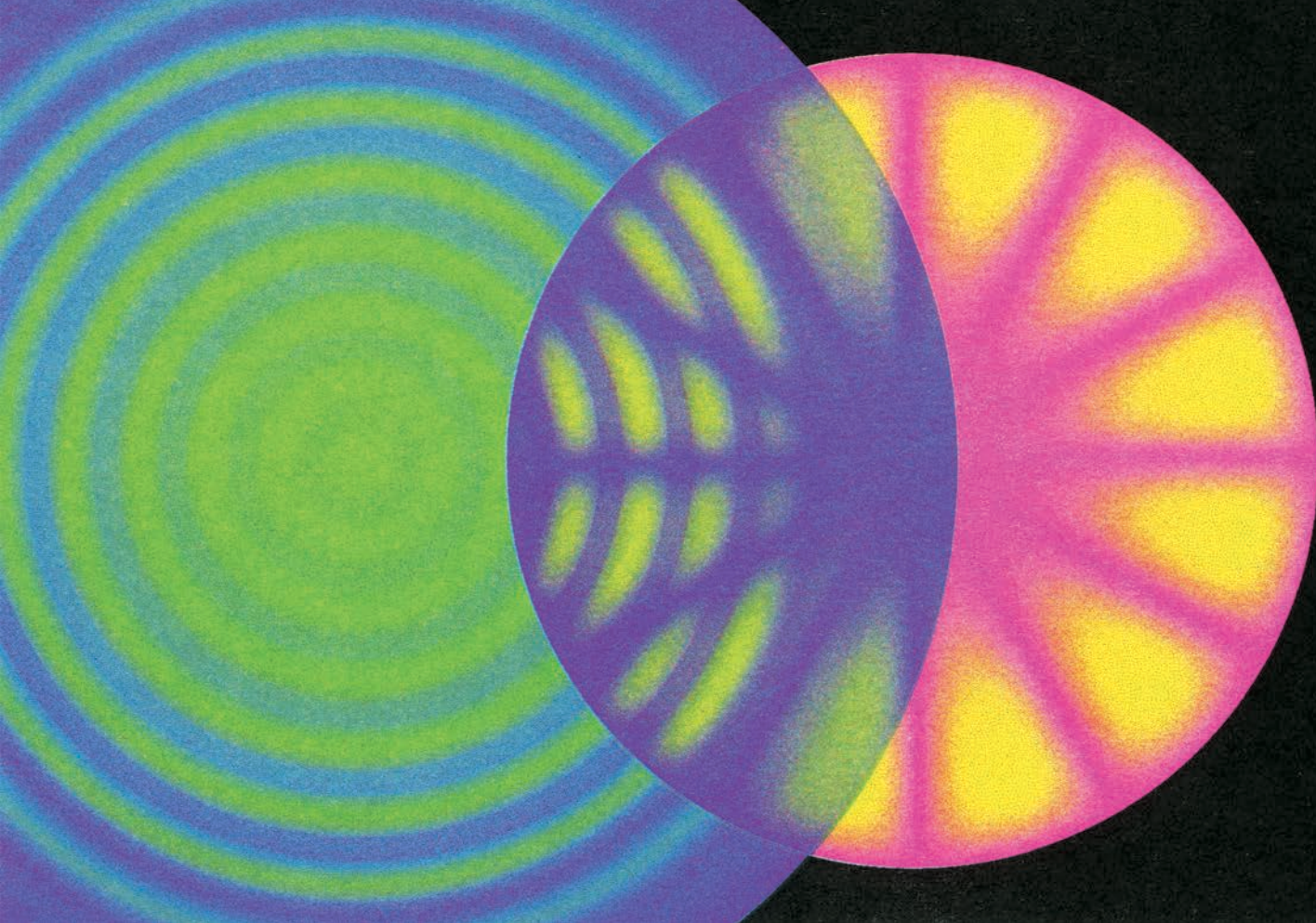
ence and guided discussions about who and what is conscious. Debates about whether it "feels like something" to be a baby, bee or bot are widely thought to carry ethical significance, marking the contrast between entities that have intrinsic moral status and those that don't.

BUT DESPITE ITS APPARENT universality, phenomenal consciousness has not always occupied center stage in how human beings have approached the mind. In fact, it is far from a universal concept. In the wonderfully named 1988 paper "Yishi, Duh, Um, and Consciousness," University of Oxford philosopher Kathleen V. Wilkes argued that there are no synonyms for "phenomenal consciousness" in ancient Greek, Mandarin, Croatian or even pre-17th-century English. (The "yishi" of Wilkes's title is perhaps the closet that Mandarin comes to a term for consciousness; "duh" and "um" are Croatian terms that have been offered as possible translations for "mind.")

Here in the Santiago region, the Indigenous language of Mapudungun addresses questions of mentality using the term "rakizum." This word captures some of what's meant by "consciousness," but it includes reflexive awareness of interdependence with other people and the natural world—relational elements that "consciousness" fails to include. One might wonder what debates about other minds might have looked like had Mapudungun, rather than English, established itself as the language of science.

The dearth of synonyms for "phenomenal consciousness" settles nothing in and of itself. (None of these languages contains synonyms for "quarks"

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or “apoptosis” either.) But it does challenge the natural assumption that understanding the space of possible minds requires appealing to the concept of consciousness. Consciousness—or at least our concept of consciousness—hasn’t always been central to discussions of the mind. Perhaps it shouldn’t be central.

SCIENCE REQUIRES CONCEPTS that support prediction, explanation and understanding. A good scientific concept is one that cuts nature “at its joints,” bundling together phenomena that share an underlying nature. The English term “consciousness” may turn out to play that role. But it’s also possible that “consciousness” will fail to make the cut—because it either bundles together phenomena that don’t really belong together or fails to bundle together things that do belong together (or both).

Precedents from this history of science are not hard to find. As philosopher Patricia Churchland once observed, the ancients grouped a wide variety of disparate phenomena together as forms of “fire”: combustion (the burning of wood); fusion (activity on the sun); electrically induced incandescence (lightning); bioluminescence (the activity of fireflies); and spectral emission (the Northern Lights). Galileo reshaped our conception of velocity by dis-

tinguishing average velocity from instantaneous velocity, and 18th-century physicist Joseph Black reshaped our understanding of heat, showing that we needed to distinguish it from temperature.

Concepts are tools. And as a tool for describing the structure of our own minds, “consciousness” is extremely useful—perhaps even indispensable. But to understand systems that are radically different from us, we may need to forge new concepts. As AI becomes increasingly sophisticated, we might find it irresistible to apply the language of consciousness to such systems, just as we are often

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tempted to use the language of consciousness in describing bees or newborn babies. But we need to recognize the limitations of consciousness talk. Certainly we should ask whether bees, babies or bots are conscious. But in asking that question, we should be alive to the possibility that the question itself might need replacing. ●